

Supplementary Information

Landscape-scale bacterial biogeography across the Rub' al-Khali

Rund Tawfiq and co-authors

4 October 2026

S1 Sample collection and processing

S1.1 Field sampling

At each site, we collected each field replicate in a sterile tube, either by scooping the sample directly into the tube or with a sterile plastic spoon. We collected samples in 2 mL tubes in T 1–T 4, and 15 mL tubes in T 5. During T 2–5, we taped the three replicate tubes together, so adjacent collections were separated by roughly one external tube diameter. Bulk subsurface soil was collected at a nominal depth of 5–10 cm; the exact sampling depth of each sample was not recorded, and for the root-adjacent compartment, the exact distance from the root was not recorded, so the compartment separations are defined operationally by how each sample was taken rather than by fixed depths or distances. Samples were held at -20°C in an Engel MR040F portable freezer during each expedition and transferred to -80°C in the laboratory.

S1.2 Amplification and indexing

The Bakt_341F (CCTACGGGNGGCWGCAG) and Bakt_785R (GACTACHVGGGTATCTAATCC) primers carried Illumina overhang adapters. First-round PCR used TaKaRa Ex Taq DNA Polymerase (catalogue RR001B) in 50 μL reactions with 5 μL of extracted DNA, 5 μL of $10\times$ Mg^{2+} -containing buffer, 4 μL of dNTP mix (2.5 mM each), 2 μL of each 10 μM primer stock, 0.25 μL of polymerase (5 U/ μL) and 31.75 μL of nuclease-free water, giving a final concentration of 0.4 μM per primer. The programme was 95°C for 3 min; 35 cycles of 95°C for 10 s, 65°C for 30 s and 72°C for 1 min; and 75°C for 5 min. We checked products on a 1 % agarose gel. Because most extracts had low or undetectable DNA, we set template input by volume: samples without a visible amplicon were re-amplified with 8 μL and then 12 μL of DNA (with 28.75 μL and 24.75 μL of water, respectively), and samples without a product after the third attempt were classified as failed.

Indexing used the QIAGEN Multiplex PCR Kit (catalogue 206143) with the following PCR conditions: 94°C for 15 min; 8 cycles of 94°C for 30 s, 55°C for 30 s and 72°C for 30 s; and 72°C for 5 min.

S1.3 Library quantification, pooling and sequencing

We purified libraries with Agencourt AMPure XP beads at a bead-suspension-to-sample volume ratio of 0.8:1, quantified them with the Qubit high-sensitivity double-stranded DNA assay, and sized them on an Agilent TapeStation 4200 with D5000 ScreenTape. We pooled libraries using the core laboratory's molar-normalization spreadsheet, whose inputs were each library's final DNA concentration and fragment size, the total pool volume and the target molarity. For a pool of N

libraries, library i with concentration C_i contributed a volume $V_i = V_{\text{pool}}C_{\text{pool}}/(NC_i)$, where V_{pool} and C_{pool} are the target pool volume and molar concentration.

S1.4 Soil pH

For analyses including soil pH measurements, we used the values frozen as EQ-PH-SHARED-v1.0.0. The v1.0.1 update retains identical measurement values and confirms that all Trip 4 pH assays used field replicate 2 because replicate 1 had been depleted. We took all readings at room temperature and recorded each only after the meter displayed its green stability indicator. Between measurements we rinsed the electrode along its full length with Milli-Q water, immersed it in a separate container of Milli-Q water, and dried it by gently tapping with Kimwipes. We calibrated the electrode at the start of each session.

The dataset contains 767 measurements, of which 712 met quality-control criteria: an unambiguous measurement date, a value within the physical range, recorded calibration, an electrode slope of 95–102 %, and a pH 10 buffer read-back of 9.9–10.1. The discarded values included 36 with ambiguous dates and 19 that failed one or more session criteria. We did not impute missing or excluded measurements. Coverage was incomplete and non-random, most notably in campaigns 1 and 3 (Table S1).

Table S1: Coverage and quality control of soil pH measurements.

Campaign	Target samples	Measured	Passed QC	Not measured	Excluded
1	330	157	157	173	0
2	27	23	23	4	0
3	450	192	156	258	36
4	177	175	156	2	19
5	234	220	220	14	0

S1.5 Soil XRF

S1.5.1 Measurement workflows

In-situ and laboratory XRF are distinct measurement workflows that differ in sample coverage, material, and analytical use, but both used the Vanta Handheld XRF Analyzer. The laboratory workflow measured stored soil samples from all campaigns on the bench, and provided 725 records. The in-situ workflow measured soil in the field during T5 only, and provided 106 measurements of which 71 (at 58 sites) passed quality control. These in-situ measurements record only the site of each reading, and do not correspond to any physical sample or compartment. Only laboratory XRF data are used for analyses, and the in-situ XRF is used as a descriptive T5 diagnostic.

The 725 records from the laboratory XRF workflow were obtained as 547 records from T1–T4 samples, and 178 records from T5. All records were obtained in Fast mode, but T5 records additionally specify the acquisition settings (Fast Screening with a helium purge and 8 mm beam, which aid detection of light elements). Where an element was measured more than once on a sample, T1–T4 records retained the largest positive reading and T5 retained the most recent reading. To assess sensitivity to the T5 reading-selection rule, we compared the last reported readings with readings marked as primary in the workbook. We found the results to be consistent for the 11 affected elements (Spearman $\rho \geq 0.994$). The combined records do not contain concentration units, limits of detection, or the meaning of non-detect. Zeros entered the $\log(1 + x)$ transformation as recorded, without a resolved detection limit interpretation.

S1.5.2 XRF element axis

We retained the 11 elements detected in at least 75 % of laboratory XRF records: Al, Ca, Cl, Fe, K, Mg, Na, P, S, Si, and Ti. We transformed their readings as $\log(1 + x)$, and standardized each element within campaign. We then reduced the standardized elements to principal components and took the first component (PC1) as a single elemental score per sample (Table S19). PC1 captured 39.2 % of the elemental variance and was weighted most strongly toward Ca (0.413), Mg (0.406), Na (0.370), S (0.354) and Cl (0.338), and negatively toward Si (-0.273). This axis contrasts higher Ca, Mg, Na, S and Cl with lower Si. This score correlated $r = 0.825$ with an internal evaporite-element reference and $r = 0.976$ with a globally standardized version of the same axis. We oriented PC1 to correlate positively with the mean within-campaign standardized log readings for Cl, Na and S, the retained elements in the internal reference set. The globally standardized sensitivity standardized each element across all campaigns together. Missing values and values constant within the standardization group were assigned zero after standardization.

S2 Controls and contamination quantification

We followed recommended guidelines for reporting contamination in low biomass samples by analyzing each control according to the study stage it tested [2]. Table S2 details the controls included in the analyses.

Table S2: Control profiles and analysis cohorts. Counts refer to sequenced profiles retained in the specified analysis; prepared-control totals are unavailable. The 24 assay-control profiles excluded from the primary survey comprise 18 extraction blanks and six PCR no-template controls (NTCs). The separate T4 rerun and mock-control analysis have their own denominators.

Control and cohort	Stage and material	Sequenced and linked analysis counts
Survey extraction blanks	Extraction; nuclease-free water and extraction reagents	18 excluded profiles; 17 mapped T5 blanks linked to 217 biological profiles
Survey PCR NTCs	Amplification; no DNA template	Six excluded profiles
T4 extraction-blank rerun	Extraction; blank controls	Six blanks linked to 95 biological profiles
T2 DNA community controls	Amplification and sequencing; ZymoBIOMICS HMW DNA Standard (D6322)	Two sequenced control libraries used for genus recovery
T3 whole-cell community controls	Extraction, amplification and sequencing; Zymo-BIOMICS Microbial Community Standard (D6300)	Two sequenced control libraries used for genus recovery

S2.1 DNA extraction kit controls

Since the study involved DNA extraction using different kits and extractions during different sessions, we included extraction blanks to understand contamination introduced during the extraction steps. We performed two sensitivity analyses using extraction blanks linked to samples from two different campaigns (T4 and T5), independently. We re-denoised the six T4 extraction blanks together with the additional control libraries and matched their ASVs to the biological count table by

exact sequence. T5 samples and their extraction blanks shared a sequencing run. For both analyses, we considered an ASV as a candidate contaminant if it was present in at least 2 blanks and was more prevalent in blanks than in samples using an unadjusted one-sided Fisher test ($p < 0.1$). We quantified the candidate contaminant ASVs in the sample profiles, then removed them and measured the effect of their removal on Shannon diversity and raw richness using Spearman’s correlation coefficient to compare the sample profiles before and after their removal.

The T5 set included 17 extraction blanks linked to 217 samples, and we found 351 candidate contaminant ASVs that comprised 2.2% of pooled reads and a median of 0.40% per profile (IQR 0.16–0.99%, maximum 56.60%). Before/after Shannon diversity and raw-richness rankings were strongly correlated ($\rho = 0.995$ and 0.99995 , respectively). T4 had 6 extraction blanks linked to 95 samples, and we found 7 candidate contaminant ASVs, including *Nesterenkonia*, *Brevibacterium* and *Alcaligenes* and 4 ASVs without a genus assignment. Candidate contaminant ASVs were present in 33% of the samples and comprised 0.14% of pooled reads (maximum 14%). Across the 95 linked T4 profiles, before/after Shannon diversity and raw-richness rankings were strongly correlated (Spearman $\rho = 0.99948$ and 1.00000 , respectively).

S2.2 Microbial community standard controls

We included two different types of mock microbial community control samples in the study design. These controls included purified DNA ZymoBIOMICS HMW DNA Standard (D6322) in T1 and T2, and whole-cell ZymoBIOMICS Microbial Community Standard (D6300) which we later extracted in T3. We used two community control replicates from T2 and two from T3 to estimate genus recovery based on the known community composition from the vendor. At the genus level, one T2 community control and two T3 controls recovered all the known genera present, with 31%, 100%, and 99.99% of their reads assigned to the expected genera, respectively.

S2.3 Well-to-well contamination

Use of 96-well plates for sequencing library preparation is known to introduce cross-contamination between physically close samples in microbiome studies [1]. We used 259 biological samples from T1 and their well positions to evaluate whether well-to-well contamination occurred in our study. Within each campaign and 96-well plate, we compared Bray–Curtis dissimilarities between pairs of samples from different sites. We defined adjacent wells as pairs of wells sharing an edge or a corner (Euclidean distance $\leq \sqrt{2}$), and we matched the pairs by geographic-distance decile and whether they came from the same or different soil compartments. For profiles listed in more than one well, we retained the first listed position. We calculated the mean difference in dissimilarity between adjacent and non-adjacent pairs, weighted by the number of adjacent pairs per stratum. The null distribution used 999 permutations of sample-to-well assignments within each plate (random seed 42). Adjacent wells were not more similar than matched non-adjacent wells. Dissimilarity difference was $+0.005$ (999 permutations, one-sided $p = 0.87$).

S3 Relic DNA experiment

S3.1 Sample description

During T5, we collected additional samples from two campsites: C1R (root-adjacent soil) and C2R, C2S (root-adjacent soil, surface soil, respectively). We also collected C2 shallow sub-surface soil, but these samples were not included in the experiments because DNA extractions failed for these

samples. For each of the three remaining parent soil tubes, representing a site $\times$ compartment group, we prepared 3 separate homogenized soil slurries in sterile phosphate-buffered saline (PBS, pH 7.4) at a 1 : 10 (w/v) ratio. We split each slurry into a treated and an untreated aliquot, giving nine paired comparisons grouped within three parent tubes.

S3.2 PMAxx treatment

We added PMAxx (Biotium, Hayward, USA) to each treated aliquot at a final concentration of 50 μ M. We subjected both treated and untreated aliquots to the following conditions: 10 min dark incubation, photo activation with a 450 nm LED lamp for 35 min on a white reflective surface (15–20 cm from the lamp), and rotation at 15 min intervals. We stored samples at -20°C until DNA extraction, which was performed using 860 μ L from each aliquot with the DNeasy PowerSoil Pro kit (QIAGEN). We amplified the DNA and sequenced the samples with the protocols described in Section S1.2.

S3.3 Results

We calculated Hurlbert expected richness at 123,897 reads, the minimum depth among the 18 libraries, and Shannon diversity at full read depth. We report each treated–untreated difference and the mean difference within each of the three site $\times$ compartment groups (Table S3, Figure S1). The headline-estimate index is under `analysis/v3/headline_uncertainty/`.

Table S3: Relic DNA aliquot comparisons for 9 sample pairs (18 libraries) across three site $\times$ compartment groups in T5 campsites.

Measure	C1R	C2R	C2S
Rarefied Richness (treated - untreated)	-36.70	-341.78	-111.49
Shannon (treated - untreated)	0.726	-0.025	-0.311

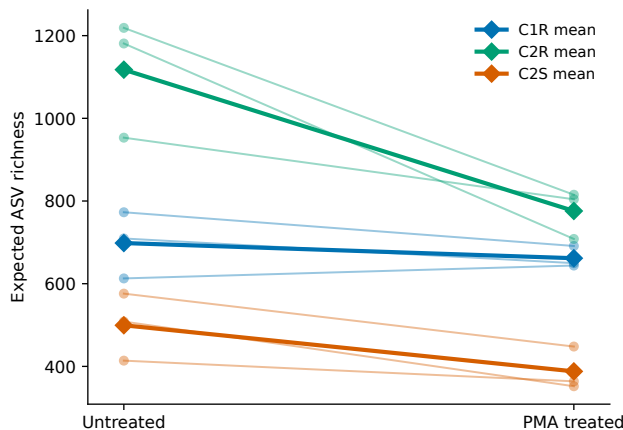


Figure S1: Expected ASV richness at 123,897 reads in untreated and PMA-treated aliquots. Thin lines join the paired aliquots from each slurry; thick lines show the mean of three slurries from each parent soil tube. C1R and C2R are root-adjacent soil at two campsites, and C2S is surface soil at the second campsite.

S4 Microbial community diversity

S4.1 Community Composition

We summed ASV counts at the phylum, class and genus levels using the SILVA v138.2 assignments and calculated the mean relative abundance of each taxon across the 1,227 core profiles, overall, by compartment and by transect third (20 sites each, ordered along the transect coordinate). Reads unassigned at a given rank were retained as an unclassified category.

Eight phyla exceeded 2% mean relative abundance (Table S4). Bacillota and Actinomycetota together accounted for 41.2% of reads, whereas Acidobacteriota and Verrucomicrobiota, which are among the most abundant phyla in soils globally, together accounted for 3.3%. Cyanobacteriota contributed 0.4% and were no more abundant in surface soil (0.40%) than in shallow-subsurface (0.41%) or root-adjacent soil (0.51%). Archaea contributed 0.02%, as expected for bacterial primers. *Domibacillus* was the most abundant genus in every compartment and in the western and central thirds, whereas *Bacillus* was the most abundant in the eastern third. Reads without a genus assignment made up 35.6% of reads. Full phylum-, class- and genus-level tables, overall and by compartment and transect third, are given in Tables S4 and S5.

Table S4: Leading phyla across the 1,227 samples. Values are mean shares of total reads in percent; prevalence is the fraction of profiles with at least one read. Compartment and transect-third columns are means within the stratum (thirds of 20 sites ordered by route position). Names follow SILVA v138.2.

Phylum	ASVs	Mean	Prev.	Surface	Shallow	Root-adj.	West	East
Pseudomonadota	88,894	26.9	100	29.9	23.8	27.0	26.3	28.2
Bacillota	48,693	22.5	100	20.8	19.6	26.2	18.3	31.5
Actinomycetota	56,581	18.7	99	18.0	20.3	18.1	22.6	12.7
Chloroflexota	24,409	8.6	99	8.5	10.7	6.9	10.3	6.0
Bacteroidota	16,657	7.1	99	8.2	5.9	7.3	6.8	6.0
Planctomycetota	20,808	3.2	98	2.3	3.5	3.8	3.1	2.9
Gemmatimonadota	12,107	2.7	98	2.9	3.5	1.8	2.7	2.8
Acidobacteriota	17,405	2.3	95	2.1	3.4	1.7	2.6	1.7
Myxococcota	15,121	1.7	98	1.6	2.0	1.5	1.7	1.4
Deinococcota	1,419	1.1	93	1.9	1.0	0.6	1.4	1.0

S4.2 Alpha diversity (within-sample)

S4.2.1 Diversity indices and rarefaction

We calculated Shannon diversity (H) with full read depths for all samples. We estimated rarefied richness as Hurlbert’s expected number of ASVs at 25,000 reads ($E[S_{25k}]$), and 55 samples were removed as they fell below the selected read depth, resulting in 1,172 samples with a richness estimate. We measured normalized Shannon as $H/\log(E[S_{25k}])$, where H is Shannon diversity and $E[S_{25k}]$ is expected richness at 25,000 reads. This mixed-depth ratio combines full-depth Shannon diversity with the logarithm of expected ASV richness at the common depth and can exceed 1. Normalized Shannon ranged from 0.336 to 0.945, and it correlated only weakly with sequencing depth (Spearman’s $\rho = 0.096$).

Table S5: Leading genera across the 1,227 samples (mean share of total reads in percent, prevalence in percent of profiles, and means by compartment and transect third).

Genus	Phylum	Mean	Prev.	Surface	Shallow	Root-adj.	West	Central	East
<i>Domibacillus</i>	Bacillota	6.22	97	6.14	4.09	8.00	7.09	6.33	4.78
<i>Massilia</i>	Pseudomonadota	3.55	94	4.85	2.14	3.66	3.69	4.58	1.97
<i>Bacillus</i>	Bacillota	2.97	98	3.82	2.64	2.58	1.56	2.11	6.22
<i>Microvirga</i>	Pseudomonadota	2.41	97	2.54	2.11	2.55	3.32	1.95	1.66
<i>Flavisolibacter</i>	Bacteroidota	2.05	94	2.88	2.00	1.44	2.49	2.27	1.10
<i>Noviherbaspirillum</i>	Pseudomonadota	1.92	94	2.62	1.76	1.50	2.22	2.10	1.23
<i>Streptomyces</i>	Actinomycetota	1.80	91	1.43	1.91	2.01	2.60	1.71	0.73
<i>Herpetosiphon</i>	Chloroflexota	1.52	92	1.68	2.07	0.95	1.45	2.19	0.72
<i>Paenibacillus</i>	Bacillota	1.38	96	1.08	1.29	1.67	1.22	1.34	1.65
<i>Lysinibacillus</i>	Bacillota	1.32	95	0.65	0.70	2.36	1.57	1.32	0.95
<i>Nocardioidea</i>	Actinomycetota	1.23	95	1.19	1.14	1.34	1.45	1.36	0.74
<i>Ammoniphilus</i>	Bacillota	1.09	96	0.89	1.46	0.95	0.53	1.10	1.92
<i>Nitribacter</i>	Bacteroidota	1.09	84	1.26	0.47	1.46	0.83	1.81	0.53
<i>Planococcus</i>	Bacillota	1.03	82	1.32	0.60	1.14	0.74	0.62	1.99
<i>Deinococcus</i>	Deinococcota	1.02	88	1.83	0.94	0.46	1.39	0.84	0.73

Table S6: Genera with the strongest monotonic change along the route among the 200 genera of the route model. ρ is the Spearman correlation of the site-level CLR value (campaign-by-compartment means removed) with route position; q is Benjamini–Hochberg over 200 tests (77 genera decreased and 47 increased eastward at $q < 0.05$). Δ is the east-minus-west difference in mean CLR between the transect thirds with a delete-one-site jackknife 95 % interval; the last two columns give the mean share of total reads in percent in the western and eastern thirds.

Genus	Phylum	ρ	q	Δ CLR [95 % CI]	West	East
<i>Strongest decreases from west to east</i>						
<i>Cellulomonas</i>	Actinomycetota	-0.79	2.3×10^{-12}	-2.23 [-2.86, -1.60]	0.50	0.13
<i>Blastocatella</i>	Acidobacteriota	-0.76	3.8×10^{-11}	-2.26 [-2.80, -1.71]	0.10	0.01
<i>Marmoricola</i>	Actinomycetota	-0.70	5.9×10^{-9}	-1.65 [-2.26, -1.05]	0.20	0.07
<i>Ellin6055</i>	Pseudomonadota	-0.70	7.9×10^{-9}	-1.71 [-2.39, -1.03]	1.12	0.26
<i>Roseisolibacter</i>	Gemmatimonadota	-0.70	9.3×10^{-9}	-1.89 [-2.49, -1.28]	0.22	0.09
<i>Steroidobacter</i>	Pseudomonadota	-0.69	1.2×10^{-8}	-1.61 [-2.10, -1.12]	0.29	0.11
<i>Limnobacter</i>	Pseudomonadota	-0.69	1.5×10^{-8}	-2.83 [-3.50, -2.17]	0.22	0.03
<i>Aridibacter</i>	Acidobacteriota	-0.69	1.5×10^{-8}	-1.86 [-2.44, -1.29]	0.08	0.02
<i>Sphingomonas</i>	Pseudomonadota	-0.67	5.5×10^{-8}	-1.35 [-1.87, -0.84]	0.21	0.14
<i>Paraconexibacter</i>	Actinomycetota	-0.67	5.5×10^{-8}	-1.84 [-2.34, -1.34]	0.05	0.03
<i>Strongest increases from west to east</i>						
<i>Halalkalibacter</i>	Bacillota	0.85	8.4×10^{-16}	4.91 [3.83, 5.99]	0.03	0.83
<i>Polygonibacillus</i>	Bacillota	0.81	6.9×10^{-13}	4.55 [3.59, 5.51]	< 0.01	0.20
<i>Ammoniphilus</i>	Bacillota	0.80	9.5×10^{-13}	1.89 [1.48, 2.31]	0.53	1.92
<i>Bacillus</i>	Bacillota	0.77	2.5×10^{-11}	1.62 [1.18, 2.07]	1.56	6.22
<i>Pseudalkalibacillus</i>	Bacillota	0.74	4.1×10^{-10}	2.89 [2.28, 3.50]	0.01	0.05
<i>Aquibacillus</i>	Bacillota	0.72	1.9×10^{-9}	4.75 [3.45, 6.04]	0.01	0.69
<i>Plastorhodobacter</i>	Pseudomonadota	0.72	3.0×10^{-9}	3.60 [2.74, 4.46]	0.02	0.10
<i>Rhodocista</i>	Pseudomonadota	0.71	5.9×10^{-9}	3.01 [2.12, 3.90]	0.03	0.20
<i>Escherichia-Shigella</i>	Pseudomonadota	0.70	7.9×10^{-9}	3.27 [2.36, 4.18]	0.01	0.84
<i>Sediminibacillus</i>	Bacillota	0.70	8.7×10^{-9}	4.59 [3.46, 5.71]	< 0.01	0.52

Table S7: Direction of the environmental gradients along the route: Spearman correlation of site means with route position and the means of the western and eastern transect thirds. Temperature is in degrees Celsius, monthly rainfall in millimetres, and humidity in percent.

Variable	Sites	ρ	West third	East third
49-month mean air temperature	60	0.98	27.89	30.07
49-month mean monthly rainfall	60	0.55	2.16	3.67
49-month mean relative humidity	60	0.92	24.72	31.82
archived-soil pH (equal-weight assay-group means within site)	60	0.82	7.85	8.23

S4.2.2 Marginal and paired compartment contrasts

We estimated compartment differences using two approaches. First, we used a marginal estimate which averages within site and compartment across all campaigns, so the two compared compartments may come from different campaigns. Second, we used a paired estimate that restricts each contrast to the campaigns that included samples from both compartments at a site, averaged the difference within campaign, then averaged the campaigns within site (Table S8). We report the paired estimate as the primary result because it removes campaign as a confound.

The two measures agree in direction throughout but differ in magnitude. The paired estimate shows lower estimated rarefied richness for root-adjacent soil in comparison to both bulk soil compartments, but in the marginal estimate the rarefied richness contrasts do not survive multiple-testing correction ($q = 0.81, 0.09, 0.07$ for root-surface, root-subsurface, and subsurface-surface). Normalized Shannon showed the most consistent signal under both estimates even after correction, where root-adjacent soil was lower than surface and shallow subsurface, and shallow subsurface was higher than surface soil. We obtained all intervals by site-level bootstrap and tested paired differences with the Wilcoxon signed-rank statistic, controlling the false discovery rate within a Shannon family and a normalized Shannon family separately.

Table S8: Campaign-matched compartment differences in three diversity measures. For each contrast we averaged the paired difference within each campaign a site was sampled, then averaged campaigns within site and the resulting site means across sites (176–182 site–campaign blocks per contrast; 60 sites). A positive value means the first-named compartment is higher. Brackets give 95% percentile site-bootstrap intervals (10^6 draws); Wilcoxon signed-rank tests assess the site-level paired differences; q controls the false discovery rate across the three contrasts within each measure.

Contrast	Mean	95% CI	q
<i>Shannon diversity (H)</i>			
Root-adjacent – Surface	−0.225	[−0.439, 0.019]	0.002
Root-adjacent – Subsurface	−0.269	[−0.492, −0.031]	0.006
Subsurface – Surface	0.144	[−0.009, 0.288]	0.010
<i>Rarefied richness (expected ASVs at 25,000 reads)</i>			
Root-adjacent – Surface	−173.2	[−300.7, −45.4]	0.023
Root-adjacent – Subsurface	−265.8	[−392.1, −139.5]	0.001
Subsurface – Surface	109.9	[9.6, 213.3]	0.115
<i>Normalized Shannon ($H/\log(E[S_{25k}])$)</i>			
Root-adjacent – Surface	−0.034	[−0.051, −0.017]	5×10^{-6}
Root-adjacent – Subsurface	−0.043	[−0.060, −0.026]	5×10^{-6}
Subsurface – Surface	0.018	[0.0073, 0.030]	0.002

S4.2.3 Potential confounds

To check whether read depth variation and DNA extraction kit differences contribute to compartment differences, we refitted the Shannon contrasts with Gaussian generalized estimating equations (GEEs) that included log read depth, campaign, and extraction kit, and used an exchangeable working correlation to account for repeated sampling of each site. The normalized Shannon contrasts, adjusted for campaign and log read depth, retained their direction and significance. A campaign $\times$ compartment interaction was significant for Shannon diversity before adjustment for read depth (Wald, $p = 0.008$) but not after ($p = 0.175$), indicating that part of the between-campaign variation in the Shannon contrasts may be attributed to read depth differences. The Shannon root-adjacent versus shallow subsurface contrast remained significant in every model we fitted for extraction kit, whereas the root-adjacent versus surface contrast in only three of the six (additive unadjusted, additive depth-adjusted, and extraction kit-matched pairs). We treat these adjustments as a sensitivity analysis rather than definitive controls, because extraction kit was recorded for 858 of the samples and was partly confounded with campaign (automated DNA extraction was used routinely from Trip 4 onward, with earlier-campaign re-extractions also processed on the instrument), so sample processing, season, and campaign cannot be fully separated.

S4.2.4 Climate associations

We first tested the correlation between our field weather measurements (air temperature, relative humidity, barometric pressure) and the three diversity indices. We averaged the measurements within site and campaign, and found that pressure correlated negatively with all diversity indices and humidity with Shannon diversity and richness, but none of the field-weather associations remained after correction when campaign and quadratic transect term were included with site-clustered standard errors. This is consistent with our assumption that these field measurements capture transient conditions rather than the climate under which microbial communities assemble.

We repeated this comparison using long-term climate data where we retrieved monthly measurements from Open-Meteo for mean air temperature, total rainfall, and mean relative humidity across the 60 sites. (Section). These covered 49 consecutive months per site, from January 2022 to January 2026 (the last month’s data only covered 1-20 January 2026). We averaged the monthly measurements per variable into single long-term means for each site that describes its typical climate over a ~ 4 year period, which resulted in mean ranges of $26.38 - 30.30^\circ\text{C}$ for temperature, $1.31 - 7.47$ mm per month for rain and $24.27 - 35.40\%$ for relative humidity. We also averaged the three alpha diversity measures (Shannon, rarefied richness, and normalized Shannon) first within site and campaign, then across campaigns. We then correlated each of the three climate means against each of the diversity means by Spearman rank across the 60 sites, resulting in nine correlations corrected together as one Benjamini-Hochberg family (Table S9). Every correlation is negative, and all effects remain after correction.

Table S9: Site-level Spearman associations between 49-month climate summaries and bacterial diversity. Brackets give 95 % paired-site percentile bootstrap intervals from 9,999 samples. The correction covers all nine tests.

Climate summary	Diversity measure	ρ [95 % interval]	q
Temperature	Shannon	-0.711 [-0.833, -0.520]	6.02×10^{-10}
	Rarefied richness	-0.795 [-0.887, -0.639]	3.12×10^{-13}
	Normalized Shannon	-0.584 [-0.753, -0.359]	1.75×10^{-6}
Rain	Shannon	-0.470 [-0.671, -0.211]	1.67×10^{-4}
	Rarefied richness	-0.495 [-0.702, -0.226]	7.51×10^{-5}
	Normalized Shannon	-0.298 [-0.532, -0.023]	0.0206
Relative humidity	Shannon	-0.701 [-0.834, -0.490]	1.04×10^{-9}
	Rarefied richness	-0.770 [-0.877, -0.593]	2.87×10^{-12}
	Normalized Shannon	-0.522 [-0.713, -0.266]	2.89×10^{-5}

S4.2.5 Effect of controls on diversity

Removing candidate contaminant ASVs from the linked T5 profiles (Section S2.1) preserved the rankings of Shannon diversity and raw ASV richness (Spearman correlation before and after of 0.995 and 0.99995, respectively); no profile fell below the rarefaction depth. Recalculating the marginal paired contrasts on the filtered sample profiles slightly changed the Shannon false discovery rate (from 0.026 to 0.031).

In a paired PMAxx experiment (Section S3) on three sample groups, *treated* – *untreated* differences in rarefied richness and Shannon varied in direction and magnitude among groups and preparations. We report these as descriptive comparisons, since the design does not support a formal test.

S4.3 Short-term rainfall

S4.3.1 Rise-and-decay distributed lag kernel

To capture the delayed biological response of microbial communities to rainfall, we modeled alpha diversity measures (Shannon, rarefied richness, and normalized Shannon) as a function of time-dependent cumulative antecedent rainfall. We averaged the replicates for each of 617 site $\times$ campaign $\times$ compartment combinations and ran the analysis independently for each of two rainfall datasets for comparison (Section). For each group, we summed rainfall over the 60 calendar days before the sample collection date under a rise-and-decay distributed lag kernel ($h(t)$) defined as:

$$h(t) = \left(\frac{t}{t_{\text{peak}}} \right) e^{1 - \frac{t}{t_{\text{peak}}}}$$

where t represents the lag time in days, and t_{peak} dictates the explicit target peak lag at which the weight reaches its maximum unit height (1.0). To isolate the rainfall pulse from confounding background signals, we adjusted for site $\times$ compartment baselines, campaign means, and campaign-specific longitude and collection day trends ($0 + C(\text{Site}) : C(\text{Compartment}) + C(\text{Campaign}) + C(\text{Campaign}) : \text{longitude}_z + C(\text{Campaign}) : \text{sampling_day}_z$). To find the optimal response window without *a priori* assumptions, we scanned 60 candidate peak lags from 0.5 to 30 days in 0.5 day increments for each of the three diversity indices, ranking candidates by the signed partial correlation, which orders models identically to the classical t statistic because they share residual degrees of freedom. We calibrated the significance of this search against a complete calendar-year

permutation distribution, described in Section S4.3.2. We assessed sensitivity to site sampling by refitting the full model, including all nuisance coefficients, over 9,999 whole-site bootstrap resamples, retaining all observations from each sampled site. We report central 95% ranges across these refits; effect-size ranges retain the observed endpoint and peak, while separate refits assess endpoint and peak reselection.

While both datasets — NASA POWER PRECTOTCORR and Open-Meteo — provided daily rainfall, the NASA POWER dataset resolved the 60 sites into 20 distinct rainfall series due to its coarse grid, while Open-Meteo resolved them more finely into 58 distinct series. Both datasets selected rarefied richness at an early lag (selected in 71.7% and 68.0% of resamples, against 25.8% and 18.3% for Shannon and 2.5% and 13.7% for normalized Shannon), agreeing on an early positive rarefied richness response but differing in magnitude and exact timing. When both endpoint and peak were reselected within a resample, the central peak ranges were 1.5–5.5 days for NASA POWER and 0.5–12.5 days for Open-Meteo. At the selected peak, each additional millimetre of peak-weighted rainfall was associated with 179.4 additional expected taxa for NASA POWER and 326.8 additional expected taxa for Open-Meteo (Table S10).

Table S10: Rainfall associations and central 95 % ranges across 9,999 whole-site refits. Every draw refits the nuisance model. Effect and partial- R^2 ranges retain the observed richness peak, and timing ranges reselect that peak. Endpoint selection searches all three diversity measures and all 60 peaks separately within each product.

Product	Taxa per mm [range]	Partial R^2 [range]	Peak, days [range]	Richness selected
NASA POWER	179.4 [63.7, 252.7]	0.036 [0.002, 0.079]	2.0 [1.5, 5.0]	71.7 %
Open-Meteo	326.8 [149.3, 797.1]	0.038 [0.005, 0.087]	1.0 [0.5, 6.0]	68.0 %

Table S11: Complete calendar-year permutation search. Each row maps the original rainfall years (2023, 2024, 2025) to the listed years for both products and all sites. Entries are the largest positive partial correlation across 60 peaks and three endpoints.

Assigned years	NASA POWER	Open-Meteo	Joint maximum
Observed assignment			
2023, 2024, 2025	0.1907	0.1956	0.1956
Permuted assignments			
2023, 2025, 2024	0.1135	0.1984	0.1984
2024, 2023, 2025	0.1859	0.0138	0.1859
2024, 2025, 2023	0.1318	0.1075	0.1318
2025, 2023, 2024	0.1898	0.0131	0.1898
2025, 2024, 2023	0.0634	0.1225	0.1225
Exact p	1/6	2/6	2/6

S4.3.2 Complete calendar-year permutation calibration

Because we selected the strongest lag, endpoint and product from a large search array, a nominal significance test could inflate the Type I error rate. To calibrate the entire search rather than the selected model, we constructed a null that breaks the correspondence between rainfall timing and

community composition while preserving the seasonal and spatial structure of the rainfall. We enumerated all six permutations of the three complete years available across every collection date in both products (2023, 2024 and 2025), where each meteorological year spans 1 December of the preceding year through 30 November. Mapping a collection date to another year preserves its month and day, so that its exposure draws on the preceding 60 days of the reassigned year. We applied the same year assignment across all sites, compartments and both products, keeping expeditions that share a year coupled, which preserves the seasonal position of sampling and the shared calendar rainfall field. The February campaign requires the preceding December and therefore excludes 2022, and the final series excludes the incomplete 2026 year. The six permutations include the observed assignment, and we computed exact one-sided probabilities as the fraction of assignments with a search statistic at least as large as observed, ties included, and this quantity has a minimum attainable value of $1/6$.

Under this null, the observed rainfall association was not distinguishable from the permuted assignments. Table S11 contains the exact probabilities, and the absolute maximum searches returned the same three values. With only three complete years available, the finite permutation orbit contains six arrangements, so $1/6$ (≈ 0.167) is the smallest probability this design can produce, and the test cannot reach conventional significance thresholds. We therefore interpret the calendar-year permutation as a coarse-resolution test under the assumption that the annual rainfall fields are exchangeable.

To confirm that the exact-ranking procedure was correctly calibrated under its own null, we applied it to 300 simulated community datasets containing site, campaign and spatially correlated campaign errors, assigning each dataset an independent, uniformly drawn year permutation. At nominal thresholds of 0.05, 0.20 and 0.50, the procedure rejected 0, 55 and 152 of the 300 datasets, respectively (exact binomial 95% intervals 0–1.22%, 14.12–23.18% and 44.86–56.46%), matching the rejection probabilities of 0, $1/6$ and $1/2$ permitted by the finite orbit. This confirms the procedure controls the error rate under the specified randomized-year null, and the field analysis additionally assumes exchangeability of the three observed annual rainfall fields.

S4.3.3 Robustness checks

To assess the sensitivity of the early-lag rarefied richness association, we ran three additional checks. First, to test whether the fitted peak was an artifact of the kernel shape, we replaced the rise-and-decay weighting with rainfall summed over seven fixed antecedent windows and repeated the endpoint search. Second, to test whether the association reflected a genuine antecedent effect rather than a spurious correlation, we constructed a negative control in which rainfall was drawn from the period *after* collection, which cannot plausibly influence the sampled community. Third, to assess sensitivity to low-biomass contaminants, we removed candidate contaminant taxa (Section S2.1) and refitted the model.

The fixed-window model placed the largest positive estimate at 3–4 days with NASA POWER and at 1 day with Open-Meteo. The post-collection negative control was weak for NASA POWER, whereas Open-Meteo gave a richness association of comparable size (maximum $t = 4.38$, partial $R^2 = 0.043$, versus 0.038 for antecedent rainfall). This product therefore supports an association around the collection date with uncertain timing. Removing candidate contaminant taxa left the antecedent estimate essentially unchanged. Allowing quadratic or cubic longitude trends reduced the NASA POWER estimate to 71–124 taxa per mm; the site-clustered interval for the quadratic-longitude model included zero (-0.15 to 141.4 taxa per mm). As an independent check on the rainfall exposure itself, we compared five rainfall datasets (CHIRPS v2.0, CMORPH V1.0, GPM IMERG Final V07B, NASA POWER and Open-Meteo) over the four complete days preceding

Trip 1 collection. Every pair was positively correlated across the 60 sites (Pearson $r = 0.501\text{--}0.917$; Spearman $\rho = 0.554\text{--}0.881$), indicating that the datasets describe a consistent spatial exposure.

S4.3.4 Wet spell attribution

To identify which “wet spells”, periods of consistent rainfall, contributed to the fitted association, we removed each regional wet spell in turn and repeated the complete search. We defined a wet spell as a maximal sequence of consecutive calendar days with recorded rainfall at any site within one campaign’s 60-day pre-collection window, and removed its rainfall jointly across all sites and compartments. This produced 29 deletions for NASA POWER and 17 for Open-Meteo.

Across these deletions, the largest positive partial correlation over the 180 models ranged from 0.0443 to 0.2030 for NASA POWER and from 0.1471 to 0.1963 for Open-Meteo. Removing the wet spell of 10–19 March 2023 produced the smallest maximum in each product, identifying it as the principal influence on the fitted early-lag association. These descriptive deletions locate the influential rainfall periods but cannot separate the biological responses to individual events; independent sampling would be required to estimate variation among responses to separate rainfall events.

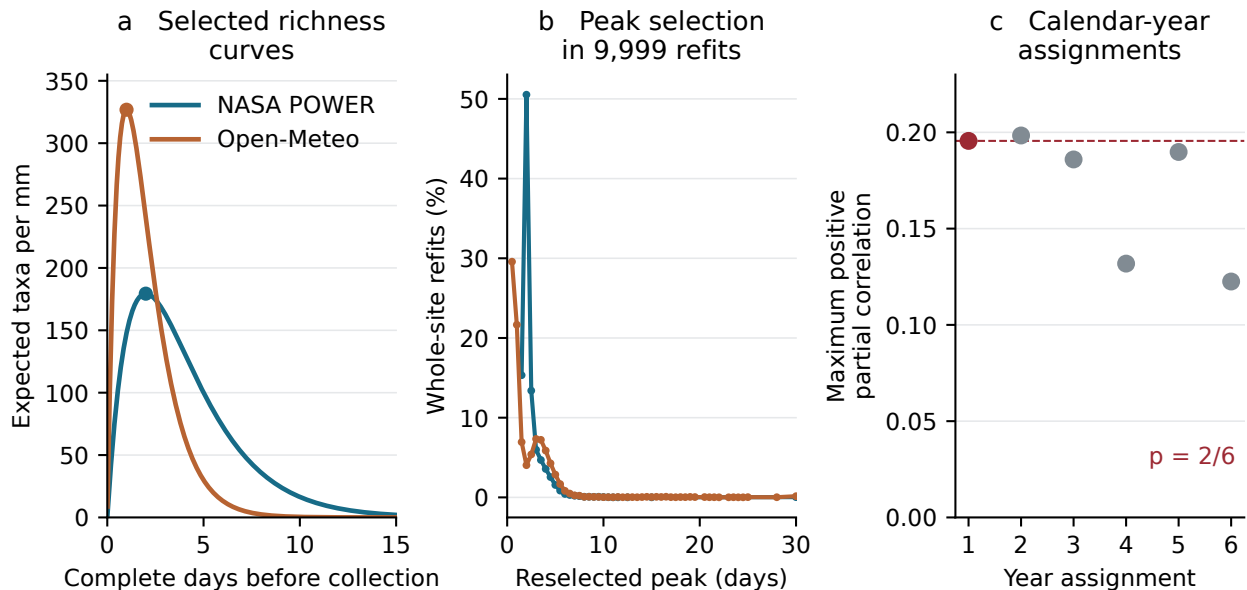


Figure S2: Rainfall associations, refitted timing distributions and complete calendar-year search. (a) Selected richness kernels for NASA POWER and Open-Meteo, expressed as expected taxa per millimetre of antecedent rain. Points mark the fitted peaks. (b) Richness-peak selection frequencies across 9,999 whole-site samples, with every nuisance model refitted. (c) The joint maximum positive partial correlation across both products, all 60 peaks and all 3 endpoints for each of the six year assignments listed in Table S11. Red marks the observed assignment, which ranked second ($p = 2/6$).

S4.4 Beta diversity (between-sample)

S4.4.1 Data preparation, resampling and multiple testing

We summed sequencing replicates within each site $\times$ campaign $\times$ compartment combination and excluded profiles with fewer than 2,000 reads, leaving 630 profiles from the 60 core sites. Campaigns

contributed 25.2, 2.5, 26.8, 27.9 and 17.5 % of these profiles. For the compositional analyses, we ranked genera present in at least 20 % of profiles by mean relative abundance (384 eligible genera) and retained the top 200 as the primary set. We CLR-transformed the raw counts of the retained genera after adding a pseudocount of 0.5, so each genus is expressed relative to the geometric mean of the retained genera. The turnover–nestedness partition and core occupancy did not use this transformation (Section S4.5).

We used the following resampling settings: 999 whole-site permutations for the transect RDA; 9,999 joint site permutations for distance decay and the turnover–nestedness regressions; 999 within-site permutations for the compartment PERMANOVA and the pH and elemental-axis partial RDAs; and 999 sign flips for all paired compartment tests. We obtained intervals for the RDA variance explained and for distance-decay slopes from delete-one-site jackknife standard errors (95 % t intervals), removing every profile or distance involving the omitted site. Intervals for compartment displacements and dispersion differences are percentile intervals from 2,000 site bootstrap resamples. We applied Benjamini–Hochberg correction within separately defined families: the three pairwise compartment location contrasts, the three dispersion contrasts, the 600 genus-level compartment contrasts, and each set of 200 genus correlations (transect, pH and elemental axis). Random seeds are recorded in the repository.

S4.4.2 Transect model

We converted site latitude and longitude to kilometres, centred them, and took the score on their first principal axis as the transect coordinate, with its sign set so that it increases eastward. The primary RDA explained 40.1 % of the variation among site profiles (delete-one-site jackknife 95 % interval 32.4–47.7 %; $p = 0.001$). We repeated the RDA with the 80, 200 and all 384 eligible genera, with linear and with quadratic trends, and after excluding each campaign in turn; all fits gave $p = 0.001$. At ASV resolution, the same model on the same 630 profiles and 60 sites explained 37.3 % of the variation with the 800 most abundant ASVs (95 % interval 32.1–42.5 %) and 35.4 % with 2,000 ASVs (30.4–40.3 %), compared with 40.1 % at genus level; all fits gave $p = 0.001$. Of the 200 genera, 124 changed monotonically along the transect ($q < 0.05$; Spearman correlation of site-level CLR values with the transect coordinate): 77 decreased and 47 increased eastward. The decreasing genera were mainly Actinomycetota (27) and Pseudomonadota (20); the increasing genera were mainly Bacillota (17) and Pseudomonadota (16) (Table S6).

S4.4.3 Residual spatial autocorrelation

The RDA residuals were spatially autocorrelated (multivariate Moran’s $I = 0.065$, $p = 0.001$, five-nearest-neighbour graph). With 3–6 neighbours, $I = 0.050$ – 0.110 ($p \leq 0.002$); with 8 and 10 neighbours, $p = 0.050$ and 0.243 , respectively. Site permutations assume that sites are independent, so we re-evaluated the transect trend under spatially correlated Gaussian error models [3]. Each model combined a site-level spatial covariance with independent profile-level noise. We varied the noise fraction (0.01, 0.10, 0.50, 0.90), the kernel (exponential, Matérn 3/2, Matérn 5/2) and its range (the 14.4-km median nearest-site distance and 5, 10, 20, 50 and 100 % of the 1,014-km transect extent). Together with a noise-only model this gave 73 models. Each model incorporated the centring and site averaging applied to the data, and we tested the trend by rotation (9,999 rotations). The trend was supported ($p = 0.0001$) at the 14.4-km range and whenever 90 % of variance was independent noise. With 50 % noise, p ranged from 0.0019 to 0.183 at ranges of 203 km or more; with ≤ 10 % noise and ranges of 50 km or more, p ranged from 0.0045 to 1.000 (Table S12). Under those models, a smooth spatial field cannot be distinguished from the transect

trend, so we report the variance explained by the transect as descriptive.

Table S12: Route rotation probabilities under the complete geometry-defined covariance family. Range is the kernel scale in kilometres; the four columns give the group-noise fraction of mean marginal site variance after centring and averaging. Each model uses 9,999 rotations and a plus-one probability. The independent-site reference and propagated group-noise-only model both give $p = 0.0001$. Monte Carlo tail intervals and covariance condition numbers are in the source TSV.

Kernel	Range (km)	1 %	10 %	50 %	90 %
Exponential	14.4	0.0001	0.0001	0.0001	0.0001
	50.7	0.1331	0.0045	0.0001	0.0001
	101.4	0.9991	0.7048	0.0001	0.0001
	202.9	1.0000	1.0000	0.0019	0.0001
	507.2	1.0000	1.0000	0.0264	0.0001
	1014.4	1.0000	1.0000	0.0500	0.0001
Matérn 3/2	14.4	0.0001	0.0001	0.0001	0.0001
	50.7	1.0000	0.6588	0.0001	0.0001
	101.4	1.0000	1.0000	0.0011	0.0001
	202.9	1.0000	1.0000	0.0457	0.0001
	507.2	1.0000	1.0000	0.1834	0.0001
	1014.4	1.0000	1.0000	0.1355	0.0001
Matérn 5/2	14.4	0.0001	0.0001	0.0001	0.0001
	50.7	1.0000	0.8979	0.0001	0.0001
	101.4	1.0000	1.0000	0.0009	0.0001
	202.9	1.0000	1.0000	0.0675	0.0001
	507.2	1.0000	1.0000	0.1831	0.0001
	1014.4	1.0000	1.0000	0.0572	0.0001

To check the error rates of these procedures, we simulated 500 null datasets per generating model on the observed coordinates and sampling design (199 rotations or permutations each). Tests using the true generating covariance rejected 3.8–5.6 % of null datasets. Unrestricted site permutation rejected 90.2 % and 100 % under two strongly autocorrelated generators. The family-maximum rotation procedure rejected none (95 % upper bound 0.74 %) (Table S15). Site permutation is therefore anti-conservative under spatial autocorrelation, and we report the RDA variance explained as descriptive.

S4.4.4 Distance decay

We restricted the analysis to sites with sample profiles in all three compartments (1,770 site pairs). Aitchison dissimilarity increased with geographic distance in every compartment (each $p = 0.0001$), at 1.349 per 100 km in surface soil (95 % CI 0.903–1.795), 1.507 in shallow-subsurface soil (1.089–1.925) and 1.175 in root-adjacent soil (0.695–1.656). The omnibus test of slope differences gave $p = 0.0051$. After maximum-statistic correction, shallow-subsurface decay exceeded root-adjacent decay by 0.332 per 100 km ($p_{\text{adj}} = 0.0039$). The difference between surface and root-adjacent soil (+0.174) was not supported ($p_{\text{adj}} = 0.173$).

S4.4.5 Compartment differences

PERMANOVA and pairwise location tests. The PERMANOVA used the 170 campaign $\times$ site blocks with all three compartments across 60 sites (pseudo- $F = 17.28$, $p = 0.001$). Each pairwise location test used every block containing both compartments of the pair: 183 blocks for shallow-subsurface versus surface, 182 for root-adjacent versus surface, and 191 for root-adjacent versus shallow-subsurface.

For each pair, we summarised how consistently compartments differed across sites as a standardized displacement:

$$\|\bar{d}\|/\|\overline{d_i}\|$$

where d_i is the site-level difference vector. This is the length of the mean difference vector divided by the mean length of the site-level difference vectors. It ranges from 0, when site-level shifts point in opposing directions and cancel, to 1, when every site shifts in the same direction. Standardized displacement was 0.331 (shallow-subsurface versus surface), 0.433 (root-adjacent versus surface) and 0.478 (root-adjacent versus shallow-subsurface), each with $p = 0.001$ and $q = 0.001$ (site bootstrap intervals in Table S13).

Table S13: Consistency of compartment shifts across 60 sites. Standardized displacement is the length of the mean within-site difference vector divided by the mean length of individual site difference vectors. Intervals are 95% percentile intervals from 2,000 whole-site bootstrap resamples. Each contrast uses its pair-specific complete site-campaign blocks.

Contrast	Blocks	Displacement	95% interval
Shallow-subsurface – surface	183	0.331	[0.306, 0.404]
Root-adjacent – surface	182	0.433	[0.386, 0.509]
Root-adjacent – shallow-subsurface	191	0.478	[0.423, 0.556]

We repeated the tests with 80, 200 and all 384 eligible genera, with pseudocounts of 0.5 and 1.0 and with multiplicative zero replacement, and separately after excluding each campaign at the primary settings. For each refit, we compared the direction of the mean difference vector with the primary analysis. All contrasts remained supported, and their directions were unchanged.

Dispersion. Dispersion increased from surface to shallow-subsurface to root-adjacent soil. Paired differences were 1.378 for shallow-subsurface minus surface (95 % CI 0.441–2.375; $q = 0.015$), 2.387 for root-adjacent minus surface (1.550–3.212; $q = 0.003$) and 1.009 for root-adjacent minus shallow-subsurface (0.056–2.055; $q = 0.061$).

Genus-level contrasts. Of the 600 genus-level tests, 295 were supported: 72 for shallow-subsurface versus surface (33 higher below the surface, 39 higher at the surface), 109 for root-adjacent versus surface (54 and 55) and 114 for root-adjacent versus shallow-subsurface (53 and 61). These differences make up the displacement vectors shown in Figure 3d (Table S14).

S4.5 Turnover, nestedness and core occupancy

We pooled raw counts of all genera within site and compartment across campaigns and rarefied each pooled community once to the smallest pooled library (12,865 reads). Mean Sørensen dissimilarity was 0.401 (surface), 0.406 (shallow-subsurface) and 0.395 (root-adjacent). The turnover component was 0.286, 0.280 and 0.295, accounting for 69–75 % of Sørensen dissimilarity. Both components increased with distance. Per 100 km, the turnover and nestedness slopes were 0.0075 and 0.0112 in surface soil, 0.0096 and 0.0164 in shallow-subsurface soil, and 0.0117 and 0.0054 in root-adjacent soil.

Table S14: Genera with the largest compartment differences: 8 largest paired mean CLR differences ($|\Delta\text{CLR}|$) in each direction for each pair of compartments. All genera shown have $q \leq 0.003$. These differences make up the displacement vectors shown in Figure 3d. Names without italics are uncultured clades without a genus name.

Genus	Phylum	$ \Delta\text{CLR} $	Genus	Phylum	$ \Delta\text{CLR} $
<i>Higher in shallow-subsurface than surface</i>			<i>Higher in surface than shallow-subsurface</i>		
<i>Polycyclovorans</i>	Pseudomonadota	1.79	<i>Planococcus</i>	Bacillota	1.75
<i>Opitutus</i>	Verrucomicrobiota	1.16	<i>Nibribacter</i>	Bacteroidota	1.48
<i>Thermosipho</i>	Thermotogota	1.02	<i>Kineococcus</i>	Actinomycetota	1.36
<i>Nordella</i>	Pseudomonadota	0.92	<i>Deinococcus</i>	Deinococcota	1.30
<i>Acidiferrimicrobium</i>	Actinomycetota	0.89	<i>Saccharibacillus</i>	Bacillota	1.27
MND1	Pseudomonadota	0.89	<i>Kineosporia</i>	Actinomycetota	1.20
I-8	Planctomycetota	0.85	<i>Rufibacter</i>	Bacteroidota	1.20
<i>Aureibacillus</i>	Bacillota	0.84	<i>Kocuria</i>	Actinomycetota	1.13
<i>Higher in root-adjacent than surface</i>			<i>Higher in surface than root-adjacent</i>		
TM7a	Patescibacteria	2.07	<i>Bradyrhizobium</i>	Pseudomonadota	2.27
<i>Pelagibacterium</i>	Pseudomonadota	1.86	<i>Deinococcus</i>	Deinococcota	2.01
<i>Arsenicitalea</i>	Pseudomonadota	1.67	<i>Oxalophagus</i>	Bacillota	1.75
<i>Ohtaekwangia</i>	Bacteroidota	1.60	<i>Blastococcus</i>	Actinomycetota	1.73
<i>Neorhizobium</i>	Pseudomonadota	1.51	<i>Vallicoccus</i>	Actinomycetota	1.64
DSSF69	Pseudomonadota	1.47	<i>Rubrobacter</i>	Actinomycetota	1.58
<i>Litchfieldia</i>	Bacillota	1.43	<i>Limnobacter</i>	Pseudomonadota	1.55
<i>Metabacillus</i>	Bacillota	1.41	<i>Cystobacter</i>	Myxococcota	1.54
<i>Higher in root-adjacent than shallow-subsurface</i>			<i>Higher in shallow-subsurface than root-adjacent</i>		
<i>Pelagibacterium</i>	Pseudomonadota	2.37	<i>Bradyrhizobium</i>	Pseudomonadota	2.18
TM7a	Patescibacteria	2.01	MND1	Pseudomonadota	2.03
<i>Planococcus</i>	Bacillota	1.99	<i>Nitrospira</i>	Nitrospirota	2.01
<i>Brevundimonas</i>	Pseudomonadota	1.95	<i>Thermosipho</i>	Thermotogota	1.99
<i>Devosia</i>	Pseudomonadota	1.95	<i>Gaiella</i>	Actinomycetota	1.94
<i>Pseudomonas</i>	Pseudomonadota	1.87	<i>Oxalophagus</i>	Bacillota	1.79
<i>Neorhizobium</i>	Pseudomonadota	1.76	<i>Symbiobacterium</i>	Bacillota	1.76
<i>Nibribacter</i>	Bacteroidota	1.75	<i>Rubrobacter</i>	Actinomycetota	1.63

Table S15: False-positive rates (%) of three tests of the transect trend, estimated from 500 simulated datasets. A correctly calibrated test rejects about 5 % at $p \leq 0.05$; with 500 datasets, the exact binomial 95 % interval is 3.3–7.3 % around 5 % and 0–0.7 % around 0 %. Site permutation: the test used for the primary RDA. True covariance: a test using the covariance that generated the data (not available for real data). Most conservative model: the largest probability across all 73 spatial covariance models. –, not evaluated.

Simulated spatial structure	Site permutation	True covariance	Most conservative model
Independent sites	5.4	4.2	0.0
Independent sample-level noise	3.6	5.4	0.0
Heavy-tailed sample-level noise	4.8	–	0.0
Exponential, 203 km range, 50 % noise	90.2	5.0	0.0
Matérn 3/2, 507 km range, 10 % noise	100.0	5.6	0.0
Range and noise not in the model set	97.8	3.8	0.0
Spatial structure varying across the transect	84.4	5.0	0.0
Different spatial structure for each genus	84.0	–	0.0

Nestedness therefore increased faster with distance in the two bulk compartments, and turnover faster in root-adjacent soil. Distance slopes and their delete-one-site jackknife intervals are given in Table S16.

Table S16: Distance decay of presence/absence dissimilarity across 60 sites (1,770 site pairs per compartment). Slopes are increases per 100 km after pooling campaigns within site and rarefying each profile to 12,865 reads. Brackets are 95% delete-one-site jackknife intervals. Two-sided probabilities use 9,999 joint site-label permutations; these are individual tests without multiplicity adjustment.

Compartment	Component	Slope [95% interval]	<i>p</i>
Surface	Sørensen	0.0187 [0.0095, 0.0279]	0.0001
Surface	Turnover	0.0075 [0.0009, 0.0140]	0.0001
Surface	Nestedness	0.0112 [0.0027, 0.0197]	0.0001
Shallow-subsurface	Sørensen	0.0261 [0.0146, 0.0375]	0.0001
Shallow-subsurface	Turnover	0.0096 [0.0024, 0.0169]	0.0001
Shallow-subsurface	Nestedness	0.0164 [0.0042, 0.0287]	0.0001
Root-adjacent	Sørensen	0.0172 [0.0106, 0.0237]	0.0001
Root-adjacent	Turnover	0.0117 [0.0033, 0.0202]	0.0001
Root-adjacent	Nestedness	0.0054 [-0.0024, 0.0133]	0.0060

To check whether differences in sequencing depth influenced the partition, we also partitioned Bray–Curtis dissimilarity on non-rarefied pooled counts. Its abundance-gradient component correlated with the absolute log ratio of library sizes ($r = 0.76\text{--}0.83$). This component is zero by construction after rarefaction, so we report it only as a diagnostic.

For core occupancy, we rarefied the same pooled counts 20 times to 12,865 reads. Root-adjacent soil had the largest core (85 genera at $\geq 90\%$ of sites, compared with 82 in surface and 69 in shallow-subsurface soil) and the highest median richness (299, 274 and 284 genera per site). 59 core genera were shared by all three compartments (Table S17).

Table S17: Genus occupancy per compartment after pooling campaigns within site and subsampling each site to 12,865 reads (20 draws; detected when present in at least half of the draws). Core genera occur at $\geq 90\%$ of sites.

Compartment	Sites	Genera detected	Core genera	Median genera per site	Mean occupancy
surface	60	912	82	274	0.281
shallow-subsurface	60	937	69	284	0.282
root-adjacent	60	1026	85	299	0.283

S4.6 Composition and environment

S4.6.1 Soil pH

709 pH measurements were linked to 560 campaign $\times$ site $\times$ compartment groups across all 60 sites (mean pH 7.99). Transect position explained 84.3 % of pH variation after accounting for campaign and compartment ($p = 0.001$). In the partial RDA, pH explained 0.278 % of the remaining compositional variation ($p = 0.125$). We partitioned variation in the adjusted site-average composition using a common response denominator: 1.38 % was uniquely associated with pH, 13.56 % uniquely with geography, 23.86 % shared and 61.20 % unexplained. Across all 560 groups, with equal total

weight per site and including within-site variation, the corresponding fractions were 0.31 %, 6.78 %, 4.76 % and 88.15 %.

For these partitions, we summed genus counts within campaign $\times$ site $\times$ compartment groups, retained groups with at least 2,000 reads, and selected the 200 most abundant genera present in at least 20 % of groups. We transformed counts to CLR values with pseudocount 0.5. Each site's groups had total weight one. Four weighted models fitted campaign-by-compartment categories alone, categories with pH, categories with linear and quadratic transect terms, and all predictors. Differences in residual sums of squares yielded the unique and shared fractions, divided by the residual sum of squares of the category-only model. For the site-average partition, we first adjusted the CLR response and each predictor against the same category design and averaged their residuals within each site; the denominator was the total variation of this fixed site-average response. These are raw descriptive fractions for two distinct responses.

We assessed sampling sensitivity with 1,999 whole-site bootstrap draws and 1,999 draws for each of three non-circular contiguous-block schemes of 5, 10 or 15 sites ordered along the transect. Each draw retained all groups from a selected site and re-estimated the category adjustment, keeping feature selection and transformation fixed. The resampling ranges describe sensitivity to the sampled sites. Removing one site at a time left the unique pH fraction at 1.25–1.63 % for site averages. The shared component ranged from 15.13 to 32.14 % across the central 95 % of whole-site bootstrap draws and from 0.22 to 31.38 % with 15-site blocks. The longer blocks span a median 253 km and retain only four drawn blocks, so the partition is sensitive to the sampled portion of the transect. Exact inputs, methods and all resampling results are provided under [analysis/v3/ph_partition_20261004/](#).

Of the 200 genera, 117 correlated with site-mean pH ($q < 0.05$), 104 of which also correlated with transect position (Table S18).

Table S18: Genera most strongly correlated with site-mean soil pH.

Increasing with pH			Decreasing with pH		
Genus	Phylum	ρ	Genus	Phylum	ρ
<i>Polygonibacillus</i>	Bacillota	0.81	<i>Caenimonas</i>	Pseudomonadota	−0.68
<i>Sediminibacillus</i>	Bacillota	0.79	<i>Pirellula</i>	Planctomycetota	−0.67
<i>Halalkalibacter</i>	Bacillota	0.79	<i>Roseisolibacter</i>	Gemmatimonadota	−0.65
<i>Halomonas</i>	Pseudomonadota	0.76	<i>Gemmatimonas</i>	Gemmatimonadota	−0.65
<i>Gracilibacillus</i>	Bacillota	0.75	<i>Steroidobacter</i>	Pseudomonadota	−0.65
<i>Aquibacillus</i>	Bacillota	0.73	<i>Streptomyces</i>	Actinomycetota	−0.64
<i>Aeromonas</i>	Pseudomonadota	0.71	<i>Gaiella</i>	Actinomycetota	−0.64
<i>Pseudalkalibacillus</i>	Bacillota	0.70	<i>Dongia</i>	Pseudomonadota	−0.63
<i>Ammoniphilus</i>	Bacillota	0.70	<i>Cellulomonas</i>	Actinomycetota	−0.60
<i>Escherichia-Shigella</i>	Pseudomonadota	0.70	<i>Sphingomonas</i>	Pseudomonadota	−0.60

S4.6.2 Laboratory XRF elemental axis

Partial RDA used 621 campaign $\times$ site $\times$ compartment profiles with a matching laboratory XRF record. In the primary analysis (200 genera, pseudocount 0.5), the elemental axis explained 0.358 % of the remaining compositional variation ($p = 0.010$). Across the 80, 200 and all 383 eligible genera in this subset and the three zero treatments, it explained 0.311–0.543 % ($p = 0.002$ –0.015). We do not report a bootstrap interval, because partial R^2 is biased upward when sites are resampled with replacement.

On Bray–Curtis dissimilarities (distance-based RDA), the axis explained 0.345 % with principal coordinates retaining 80 % of variation (38 axes, $p = 0.056$) and 0.330 % at 95 % ($p = 0.044$). The

effect size is therefore similar across geometries, while its significance is close to 0.05. Site-mean residuals remained spatially autocorrelated after accounting for campaign, compartment and the elemental axis (multivariate Moran’s $I = 0.304$, $p = 0.001$). 118 genera correlated with the axis ($q < 0.05$; Table S20).

Table S19: Element loadings on the laboratory-XRF elemental axis (PC1 after within-campaign standardization).

Element	Ca	Mg	Na	S	Cl	Fe	Ti	K	Al	P	Si
Loading	0.41	0.41	0.37	0.35	0.34	0.28	0.28	0.17	0.16	-0.01	-0.27

Table S20: Genera whose site-level CLR values correlate with the site-mean laboratory XRF axis (Spearman, 60 sites). 10 with the largest $|\rho|$ in each direction are shown. Positive ρ indicates higher abundance towards the Ca, Mg, Na, S and Cl end of the axis, negative ρ towards the Si end.

Genus	Phylum	ρ	q
<i>Increasing along the axis</i>			
<i>Polygonibacillus</i>	Bacillota	0.81	1.8×10^{-12}
<i>Escherichia-Shigella</i>	Pseudomonadota	0.74	6.2×10^{-10}
<i>Sediminibacillus</i>	Bacillota	0.74	6.2×10^{-10}
<i>Aquibacillus</i>	Bacillota	0.74	6.2×10^{-10}
<i>Halalkalibacter</i>	Bacillota	0.73	1.2×10^{-9}
<i>Gracilibacillus</i>	Bacillota	0.72	3.9×10^{-9}
<i>Enterococcus</i>	Bacillota	0.71	5.4×10^{-9}
<i>Aeromonas</i>	Pseudomonadota	0.69	1.7×10^{-8}
<i>Pseudalkalibacillus</i>	Bacillota	0.69	2.5×10^{-8}
<i>Crossiella</i>	Actinomycetota	0.65	2.3×10^{-7}
<i>Decreasing along the axis</i>			
<i>Steroidobacter</i>	Pseudomonadota	-0.71	5.4×10^{-9}
<i>Caldilinea</i>	Chloroflexota	-0.67	9.3×10^{-8}
<i>Phenylobacterium</i>	Pseudomonadota	-0.65	2.7×10^{-7}
<i>Paraconexibacter</i>	Actinomycetota	-0.65	3.0×10^{-7}
<i>Streptomyces</i>	Actinomycetota	-0.65	3.1×10^{-7}
<i>Roseisolibacter</i>	Gemmatimonadota	-0.63	7.1×10^{-7}
<i>Gemmatimonas</i>	Gemmatimonadota	-0.63	7.2×10^{-7}
<i>Pirellula</i>	Planctomycetota	-0.58	8.9×10^{-6}
<i>Blastocatella</i>	Acidobacteriota	-0.58	8.9×10^{-6}
<i>Dongia</i>	Pseudomonadota	-0.58	8.9×10^{-6}

S4.6.3 Landform

Each site has one assigned landform: 44 sand dunes, 9 saline pans (three western, one central, five eastern), 3 eastern oases, and one each of gravel plain, dune slack, aeolian lake and oil-spill site (Table S21).

Alpha diversity. Dune and saline pan sites did not differ in Shannon diversity, rarefied richness or normalized Shannon diversity according to Mann–Whitney tests, which gave $p = 0.35$, 0.28 and 0.93, respectively (Table S21). The three oasis sites had the lowest diversity (mean Shannon 4.25, compared with 5.77 at dune sites), which we report descriptively as there are only three such sites.

Composition. Saline pans differed from sand dunes before adjustment for transect position (pseudo- $F = 4.56$, $p = 0.0013$), with 16 genera differing at $q < 0.05$. Examples of genera more

abundant in sand dunes were *Herpetosiphon*, *Oxalophagus*, *Opitutus* and *Symbiobacterium*, examples of genera more abundant in saline pans were *Thermobaculum*, *Truepera* and *Enterococcus*. After adjustment, the difference was smaller (pseudo- $F = 1.78$, $p = 0.032$) and no genus passed $q < 0.05$ (Table S22).

Dune-only refits. On the 44 dune sites, the transect RDA explained 34.2 % of compositional variation (compared with 40.1 % across all sites), and 70 of the 71 genera with supported transect correlations changed in the same direction as in the full dataset. Of the nine climate–diversity correlations, the six involving temperature and humidity and the rainfall–rarefied richness correlation remained supported ($q < 0.05$); rainfall–Shannon ($q = 0.141$) and rainfall–normalized Shannon ($q = 0.459$) did not (Table S23).

Table S21: Landform of the 60 sites and site-mean diversity by landform.

Landform	Sites	Thirds	Shannon	Rarefied Richness	Normalized Shannon
sand dune	44	15 west, 18 central, 11 east	5.77	1635	0.806
saline pan	9	3 west, 1 central, 5 east	5.15	1259	0.788
desert oasis	3	3 east	4.25	578	0.737
aeolian lake	1	1 east	4.44	759	0.783
dune slack	1	1 west	6.08	1825	0.815
gravel	1	1 west	5.45	1496	0.799
oil spill	1	1 central	6.21	2291	0.811

Table S22: Landform sensitivity of the transect model.

A. Dune vs saline pan			
Adjustment	Pseudo- F	p	Genera at $q < 0.05$
None	4.56	0.0013	16
Transect position	1.78	0.032	0
B. Transect model			
Sites	R^2	Transect genera at $q < 0.05$	
All ($n = 60$)	0.401	124	
Dune ($n = 44$)	0.342	71	

S5 Functional potential

S5.1 Prediction input and quality

We generated PICRUSt2 v2.4.1 predictions from two input tables (Trips 1–4 and Trip 5 ampliseq outputs; the Trip 5 input contained 330,830 ASVs), and all 1,227 samples were matched to a prediction. The weighted NSTI had a median of 0.165 across sample profiles (range 0.005–0.782), with medians of 0.201 in shallow-subsurface, 0.168 in surface and 0.134 in root-adjacent profiles. The execution parameters, including hidden-state method, placement settings and NSTI cut-off, are archived with the analysis.

Table S23: Spearman correlations between site climate and diversity, across all sites and across dune sites only. q values are BH adjusted.

Climate variable	Diversity measure	All sites ($n = 60$)		Dune sites ($n = 44$)	
		ρ	q	ρ	q
Temperature	Shannon	-0.71	6.0×10^{-10}	-0.57	1.4×10^{-4}
	Rarefied richness	-0.79	3.1×10^{-13}	-0.71	5.5×10^{-7}
	Normalized Shannon	-0.58	1.8×10^{-6}	-0.44	4.5×10^{-3}
Humidity	Shannon	-0.70	1.0×10^{-9}	-0.57	1.4×10^{-4}
	Rarefied richness	-0.77	2.9×10^{-12}	-0.69	7.6×10^{-7}
	Normalized Shannon	-0.52	2.9×10^{-5}	-0.38	0.017
Rainfall	Shannon	-0.47	1.7×10^{-4}	-0.23	0.14
	Rarefied richness	-0.49	7.5×10^{-5}	-0.32	0.044
	Normalized Shannon	-0.30	0.021	-0.11	0.46

S5.2 Comparison with metagenomes

Shotgun sequencing of 150 Trip 5 libraries (27 surface, 44 shallow-subsurface and 79 root-adjacent) yielded a catalogue of 991 metagenome-assembled genomes (MAGs). We estimated genome abundances in each shotgun library with CoverM and annotated the catalogue using eggNOG-mapper v2.1.12. On average, 28.5 % of shotgun reads mapped to the catalogue, so the MAG-derived community profiles summarize the recovered fraction of each library.

For the whole-profile comparison, we used 990 genomes with KO annotations to construct an abundance-weighted community KO profile for each shotgun library. We compared the 125 libraries with matched amplicon profiles across 3,471 shared KOs. The marker-gene comparison used the intersection of 975 genomes with the required annotation and abundance data and 119 shotgun libraries with matched amplicon profiles. We correlated predicted and MAG-derived community abundances for each of the 20 marker-gene sets, with confidence intervals from 9,999 bootstrap resamples of sites that kept libraries from the same site together.

S5.3 Pathway composition

Of the 462 predicted pathways, 416 were present in at least 20 % of the 633 grouped sample profiles. We repeated the transect and compartment analyses, first with the 100 most abundant pathways, then with all 416 prevalent pathways. We added a pseudocount of 1.0 and excluded each campaign in turn. Unlike the community composition analysis, the pairwise compartment tests for pathways were computed within the 172 complete three compartment blocks, each centred on its three compartment mean, rather than within pair-specific blocks. Intervals for standardized displacement came from 2,000 site-level bootstrap resamples.

We assigned pathways to classes by matching keywords in their MetaCyc descriptions: biosynthesis (*biosynthesis*), degradation or utilization (*degradation*, *utilization*, *assimilation*), energy and central metabolism (*respiration*, *fermentation*, *glycolysis*, *TCA cycle*, and *carbon-fixation cycles*), and other. These predicted pathway abundances are not independent as pathways can share gene families.

S5.4 Marker-gene sets

analysis/v3/trait_genes_20260909/trait_definitions.tsv lists the KOs in each of the 20 marker-gene sets. The hydrogenase set combines *hyaB*, *hhyL* and *hozH*, and therefore includes

both membrane-bound group 1 and soluble group 3d [NiFe]-hydrogenases. Because a genome counts as a carrier when it contains any KO in a set, multi-gene sets (for example, osmolyte synthesis) indicate partial rather than complete pathways. We calculated community carrier abundance as the summed relative abundance of carrier genomes among the genomes recruiting reads in each library. Genome-level results are shown in Table S25 and PICRUST2 results in Table S26.

Table S24: Leading supported predicted-pathway differences between compartments: the five largest paired mean CLR differences ($|\Delta\text{CLR}|$) in each direction among the 270 supported tests ($q < 0.05$) of the 600-test family. MetaCyc identifiers are given in the archived results.

Pathway	$ \Delta\text{CLR} $	Pathway	$ \Delta\text{CLR} $
<i>Higher in subsurface than surface</i>		<i>Higher in surface than subsurface</i>	
L-isoleucine biosynthesis IV	0.22	L-alanine biosynthesis (superpathway)	0.36
Incomplete reductive TCA cycle	0.15	tRNA processing	0.35
Adenine and adenosine salvage III	0.14	8-amino-7-oxononanoate biosynthesis I	0.33
Geranylgeranyl diphosphate biosynthesis II	0.12	Biotin biosynthesis I	0.31
GDP-mannose biosynthesis	0.09	TCA cycle VII (acetate producers)	0.24
<i>Higher in root-adjacent than shallow-subsurface</i>		<i>Higher in shallow-subsurface than root-adjacent</i>	
Formaldehyde assimilation II (RuMP cycle)	0.53	L-isoleucine biosynthesis IV	0.38
Palmitate biosynthesis II	0.49	GDP-mannose biosynthesis	0.19
L-lysine biosynthesis II	0.37	Kdo transfer to lipid IV _A III	0.19
TCA cycle VII (acetate producers)	0.35	Glycogen degradation I	0.17
L-alanine biosynthesis (superpathway)	0.31	Purine nucleotide <i>de novo</i> biosynthesis II	0.17
<i>Higher in root-adjacent than surface</i>		<i>Higher in surface than root-adjacent</i>	
Formaldehyde assimilation II (RuMP cycle)	0.51	Kdo transfer to lipid IV _A III	0.17
1,4-dihydroxy-2-naphthoate biosynthesis I	0.31	L-isoleucine biosynthesis IV	0.16
Phylloquinol biosynthesis (superpathway)	0.30	Starch degradation V	0.16
L-lysine biosynthesis II	0.28	CMP-Kdo biosynthesis I	0.15
Menaquinol-11 biosynthesis (superpathway)	0.27	tRNA processing	0.14

Table S25: Marker-gene sets in the 975 annotated MAGs of the companion catalogue. Genomes: percentage carrying at least one KO in the set. Su, Sh, Ro: mean relative abundance of carrier genomes in surface, shallow-subsurface and root-adjacent libraries. Leading carriers: phylum with the most carrier genomes (number of genomes); Act., Actinomycetota; Pse., Pseudomonadota; Bac., Bacillota. KO memberships are archived with the analysis.

Marker set	Genomes (%)	Su	Sh	Ro	Leading carriers
CO dehydrogenase (<i>coxL</i>)	38.5	43.8	45.5	40.4	Act. (170)
[NiFe]-hydrogenase (<i>hyaB</i> , <i>hhyL</i> , <i>hoxH</i>)	22.6	17.2	31.9	15.0	Act. (126)
RuBisCO (<i>rbcL/cbbL</i>)	24.6	18.2	24.1	18.1	Act. (90)
Photosystem II D1 (<i>psbA</i>)	0.2	0.1	0.2	0.1	Pse. (1)
Nitrogenase (<i>nifH</i>)	1.0	0.5	0.3	0.8	Pse. (3)
Ammonia/methane monooxygenase (<i>amoA/pmoA</i>)	3.3	3.3	4.7	1.9	Act. (16)
Nitrite reductase (<i>nirK</i> , <i>nirS</i>)	32.4	30.6	38.8	23.1	Act. (149)
Nitrous-oxide reductase (<i>nosZ</i>)	4.1	5.1	4.0	2.2	Pse. (14)
Urease (<i>ureC</i>)	31.2	24.0	27.9	35.3	Pse. (132)
Trehalose, OtsAB (<i>otsA</i> , <i>otsB</i>)	72.7	85.6	89.0	83.8	Act. (336)
Trehalose, TreYZ (<i>treY</i> , <i>treZ</i>)	70.1	74.3	79.1	76.5	Act. (258)
Ectoine (<i>ectA-ectC</i>)	36.4	25.9	28.0	31.3	Act. (145)
Carotenoids (<i>crtB</i>)	49.3	60.6	52.7	53.4	Pse. (199)
Sporulation (<i>spo0A</i>)	7.3	5.1	4.3	7.5	Bac. (41)
Photolyase (<i>phrB</i>)	23.6	34.5	17.8	26.0	Pse. (71)
Excinuclease A (<i>uvrA</i>)	94.6	93.2	92.0	93.8	Act. (324)
Recombinase A (<i>recA</i>)	92.2	88.9	90.9	92.4	Act. (324)
Catalase (<i>katE</i> , <i>katG</i>)	77.4	86.7	76.3	84.1	Act. (266)
Superoxide dismutase (<i>sodA</i>)	82.8	68.4	67.4	78.8	Pse. (238)
Dps (<i>dps</i>)	65.1	58.2	47.3	69.5	Act. (177)

Table S26: PICRUST2 predictions for the marker-gene sets. Fold changes are paired site-level ratios of the predicted share of KO abundance between compartments (Su, surface; Sh, shallow-subsurface; Ro, root-adjacent; sign-flip tests, Benjamini–Hochberg over 60 tests). Transect ρ : Spearman correlation of site means with the transect coordinate (Benjamini–Hochberg over 20 tests). Agreement ρ : Spearman correlation between predicted and genome-derived shares across 119 matched libraries. * $q < 0.05$. Predicted shares by compartment are archived with the analysis.

Marker set	Sh/Su	Ro/Su	Ro/Sh	Transect ρ	Agreement ρ
CO dehydrogenase (<i>coxL</i>)	1.09*	0.78*	0.74*	−0.11	0.71
[NiFe]-hydrogenase (<i>hyaB</i> , <i>hhyL</i> , <i>hoxH</i>)	1.16*	0.77*	0.67*	0.05	0.74
RuBisCO (<i>rbcL/cbbL</i>)	1.01	0.81*	0.82*	0.26	0.33
Photosystem II D1 (<i>psbA</i>)	1.25	0.90	0.81	0.05	0.18
Nitrogenase (<i>nifH</i>)	1.19	1.19	1.04	0.08	0.04
Ammonia/methane monooxygenase (<i>amoA/pmoA</i>)	1.33	1.36	1.21	−0.24	0.41
Nitrite reductase (<i>nirK</i> , <i>nirS</i>)	0.96	0.88*	0.93	−0.23	0.40
Nitrous-oxide reductase (<i>nosZ</i>)	1.01	0.95	0.95	0.20	−0.10
Urease (<i>ureC</i>)	0.92*	0.98	1.08*	−0.27	0.26
Trehalose, OtsAB (<i>otsA</i> , <i>otsB</i>)	1.01	0.86*	0.87*	−0.71*	0.53
Trehalose, TreYZ (<i>treY</i> , <i>treZ</i>)	0.95	0.84*	0.91	−0.67*	0.40
Ectoine (<i>ectA-ectC</i>)	1.21*	1.22*	1.05	0.36*	0.65
Carotenoids (<i>crtB</i>)	0.97	0.92*	0.94*	−0.59*	0.13
Sporulation (<i>spo0A</i>)	0.91	1.41*	1.49*	0.56*	0.51
Photolyase (<i>phrB</i>)	1.03	0.94*	0.93*	−0.27	−0.14
Excinuclease A (<i>uvrA</i>)	1.03*	0.99	0.96*	−0.24	−0.21
Recombinase A (<i>recA</i>)	1.04*	1.00	0.95*	−0.57*	0.06
Catalase (<i>katE</i> , <i>katG</i>)	0.91*	1.09*	1.20*	−0.16	0.57
Superoxide dismutase (<i>sodA</i>)	1.00	1.06*	1.06*	0.70*	0.66
Dps (<i>dps</i>)	0.97*	1.07*	1.10*	0.43*	0.76

Table S27: Predicted MetaCyc pathways most strongly correlated with transect position (Spearman ρ between site-level CLR values and the transect coordinate; 60 sites). Of 200 pathways tested, 92 were significant (Benjamini–Hochberg $q < 0.05$): 66 increased and 26 decreased eastward. All pathways shown have $q < 0.001$.

Increasing eastward		Decreasing eastward	
Pathway	ρ	Pathway	ρ
Queuosine biosynthesis	0.75	<i>Bifidobacterium</i> shunt	−0.72
Pantothenate and coenzyme A biosynthesis I	0.73	Glycolysis and Entner–Doudoroff (superpathway)	−0.63
tRNA charging	0.67	Galactose degradation I (Leloir pathway)	−0.59
Phosphopantothenate biosynthesis I	0.67	UDP-glucose-derived O-antigen building blocks (superpathway)	−0.56
Pyrimidine deoxyribonucleotide <i>de novo</i> biosynthesis I	0.65	Purine nucleotide degradation II (aerobic)	−0.56
preQ ₀ biosynthesis	0.63	TCA cycle VI (obligate autotrophs)	−0.53
6-Hydroxymethyl-dihydropterin diphosphate biosynthesis I	0.63	Palmitate biosynthesis II	−0.53
Anhydromuropeptide recycling	0.62	L-Glutamate and L-glutamine biosynthesis	−0.52
Gondoate biosynthesis (anaerobic)	0.62	Fatty acid β -oxidation I	−0.48
L-Threonine biosynthesis (superpathway)	0.61	Fatty acid salvage	−0.48

References

- [1] Caitriona Brennan, Pedro Belda-Ferre, Simone Zuffa, Vincent Charron-Lamoureux, Ipsita Mohanty, Gail Ackermann, Celeste Allaband, Madison Ambre, Tara Boyer, MacKenzie Bryant, Kalen Cantrell, Antonio Gonzalez, Daniel McDonald, Rodolfo A. Salido, Se Jin Song, Gillian Wright, Pieter C. Dorrestein, and Rob Knight. Clearing the plate: a strategic approach to mitigate well-to-well contamination in large-scale microbiome studies. *mSystems*, 9(10):e00985–24, 2024.
- [2] Noah Fierer, Pok Man Leung, Rachael Lappan, Raphael Eisenhofer, Francesco Ricci, Sophie I. Holland, et al. Guidelines for preventing and reporting contamination in low-biomass microbiome studies. *Nature Microbiology*, 10(7):1570–1580, 2025.
- [3] Øyvind Langsrud. Rotation tests. *Statistics and computing*, 15(1):53–60, 2005.